

Clinical Risk Assessment Report: Patient 34

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.47x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.4%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -1.12 [Bottom 13%] (-2.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.02 [Avg (74th %ile)] (-0.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.59 [Bottom 24%] (-0.9%)**
Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Carcinoembryonic Antigen (CEA) **Value: 1.48 [Bottom 15%] (-0.9%)**
Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.02 [Top 22%] (-0.7%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.5 [Avg (35th %ile)] (-0.6%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.77 [Bottom 4%] (-0.5%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -1.62 [Bottom 9%] (-0.4%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)

